Lab-2 Assignment

Procedure

As in Lab 1, accept your assignment, create a new project and put the project under revision control.

Background

We have been introduced to a number of new assembly instructions:

- BL - Branch with Link (Call a subroutine) and store the return address in the LR 2. B - Branch
- BX LR - Return from a subroutine.
- PUSH Reg.. - Push multiple Registers to the stack
- POP Reg.. - Pop multiple Registers from the stack.
- LDR Rt, label - Load register from memory LDR, PC-relative

Phase 1

- From last week, start a project.
- Load any three general purpose registers with 0x00001111 and 0x00002222 and 0x00003333. Push these values to the stack one at a time using the PUSH command. Observe the stackpointer as you do this. Is the stack pointer going up or down?
- Open a memory window under View ->. Open the location that the Stack Pointer is at. Look at the memory values and how they contrast to the register values.
- Pop these registers off the stack and observe the Stack Pointer.
- Repeat the process but use multiple PUSH and POP’s on one line.

Phase2

Now continue to build your program that has the following features:

- Store an integer in a register.
- Create a subroutine that will calculate the factorial of this integer and store it in a register. After returning from this subroutine, demonstrate you have calculated the factorial by debugging and investigate the register.

Phase3

Now continue to build your program that has the following features:

- Store two strings ”ENSE 352 is fun and I am learning ARM assembly!” and ”Yes I really love it!”
- Your program will count the vowels of these two strings and store that count in a register that is available in the main part of your program.
- Your program shall have a subroutine named countVowels.
- Your program shall count both upper case and lower case vowels as the same vowel.

Submission

Make sure you have commited your latest solution and pushed it to your repository for grading.